Algebra 2 Linear Functions & Systems Cheat Sheet

Unit Focus: *slope, forms of a line, linear models, and best-fit lines*

1 KEY VOCABULARY

slope	The steepness of a line: $m = \frac{\text{rise}}{\text{run}} = \frac{y \text{ change}}{x \text{ change}}$. Rise over run, **never** run over rise.
rate of change	How fast one quantity changes per unit of another, with units (dollars per hour). For a line it is the slope.
***y*-intercept**	The point $(0, b)$ where the line crosses the y-axis: the value when $x = 0$ (the starting amount).
slope-intercept form	$y = mx + b$. Shows the slope and y-intercept at a glance.
point-slope form	$y - y_1 = m(x - x_1)$. Built from one point and the slope.
standard form	$Ax + By = C$, with integers A, B, C and $A \geq 0$.
parallel lines	Lines in a plane that never meet: **equal slopes**, different y-intercepts.
perpendicular lines	Lines that meet at a right angle: slopes are **negative reciprocals**, $m_1 \cdot m_2 = -1$.
residual	Actual y − predicted y for one data point. Positive means the line is too low there.
correlation (*r*)	Measures linear fit, $-1 \leq r \leq 1$. Near ±1 is strong; near 0 is weak. It does **not** prove cause.
feasible region	The set of points that satisfy **every** constraint of a system of inequalities. Its corners are the vertices.

2 SLOPE & RATE OF CHANGE

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y \text{ change}}{x \text{ change}}$$

Example: Slope through (−2, 5) and (4, −7)

$m = \frac{-7 - 5}{4 - (-2)} = \frac{-12}{6} = \mathbf{-2}$ (subtract in the **same order** top and bottom)

From a table: x: 2, 5, 8 and y: 7, 19, 31.
Rates $\frac{19 - 7}{5 - 2} = 4$ and $\frac{31 - 19}{8 - 5} = 4$, so it is **linear** (constant rate).

***m* = 0:** horizontal, $y = k$

***m* undefined:** vertical, $x = h$

In context: the slope is the change in y per unit of x, with units.

3 THREE FORMS OF A LINE

Slope-intercept	$y = mx + b$ · best for graphing and modeling (b is the start)
Point-slope	$y - y_1 = m(x - x_1)$ · best with a point and a slope
Standard	$Ax + By = C$ · slope $-\frac{A}{B}$; intercepts from $x = 0$ or $y = 0$

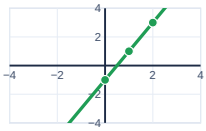
Example: Line through (−2, 5) and (4, −7); *m* = −2

Point-slope with (−2, 5): $y - 5 = -2(x + 2)$ **(sign flip: $x - (-2) = x + 2$)**
Slope-intercept: $y = -2x + 1$ Standard: $\mathbf{2x + y = 1}$
Check (4, −7): $2(4) + (-7) = 1$ ✓

Find *b*: substitute a point. Vertical lines, $x = h$, have no slope-intercept form.

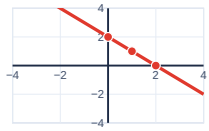
4 LINE GALLERY: SLOPE & RELATIONSHIPS

Positive slope



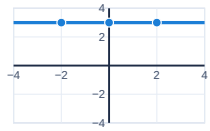
$y = 2x - 1$
m = 2: rises left to right

Negative slope



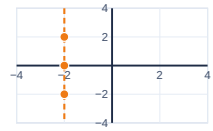
$y = -x + 2$
m = −1: falls left to right

Zero slope



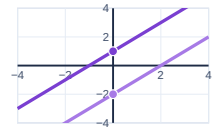
$y = 3$
m = 0: horizontal

Undefined slope



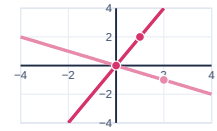
$x = -2$
m undefined: vertical

Parallel



$y = x + 1$
$y = x - 2$
same ***m***, different **b**

Perpendicular



$y = 2x$ $y = -\frac{1}{2}x$
$m_1 \cdot m_2 = -1$

Dots mark lattice points: rises = positive, falls = negative, flat = 0, vertical = undefined.

5 PARALLEL & PERPENDICULAR LINES

PARALLEL
$m_1 = m_2$

PERPENDICULAR
$m_2 = -\frac{1}{m_1}$ (flip **and** change sign)

Example: Line $y = \frac{2}{3}x - 1$. Write lines through (6, 1).

Parallel: $m = \frac{2}{3}$. $y - 1 = \frac{2}{3}(x - 6) \rightarrow$ $y = \frac{2}{3}x - 3$

Perpendicular: $m = -\frac{3}{2}$. $y - 1 = -\frac{3}{2}(x - 6) \rightarrow$ $y = -\frac{3}{2}x + 10$

$3x + 4y = 8$ has slope $-\frac{3}{4}$; perpendicular slope: $\frac{4}{3}$.

6 LINEAR MODELS & BEST-FIT LINES

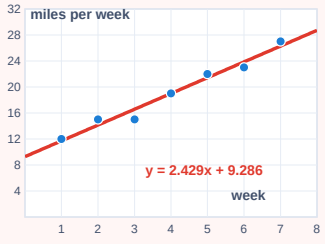
Example: Weekly miles run, weeks 1–7: 12, 15, 15, 19, 22, 23, 27

Regression: $\mathbf{y = 2.429x + 9.286}$, $r \approx 0.986$ (strong)

Slope: about 2.4 more miles each week.
Intercept: about 9.3 miles at week 0 (the start).

Residual at week 3: 15 − (2.429·3 + 9.286) = **−1.57** (line is too high)

Interpolate: week 4.5 gives about 20.2 miles. **Extrapolate:** week 20 gives 57.9 miles, unreliable!



7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:
- Slope is **rise over run**: change in y on top.
- Perpendicular slope: flip the fraction **and** change the sign.
- In point-slope form, $x - (-2)$ becomes $x + 2$.
- Give units when you interpret a slope or intercept.

COMMON MISTAKES:
- Run over rise, or subtracting the points in different orders.
- Flipping the perpendicular slope but not changing its sign.
- Calling the y-intercept "meaningless" when it is the starting amount.
- Trusting a model far outside the data (extrapolating).

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More Practice: *inequalities, systems, linear programming, and three variables*

1 LINEAR INEQUALITIES IN TWO VARIABLES

Example: Graph $3x - 2y > 6$

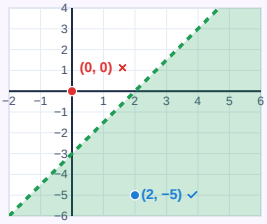
1. Solve for y: $-2y > -3x + 6$

2. Divide by -2 and **flip**: $y < \frac{3}{2}x - 3$

3. Boundary $y = \frac{3}{2}x - 3$: **dashed** ($>$ has no "equal to")

4. Test (0, 0): $0 > 6$ is **false**, so shade the side **without** the origin (below).

$(2, -5)$: $6 + 10 = 16 > 6$ ✓



< > dashed **≤ ≥** solid

$y >$ or $\geq$: above $y <$ or $\leq$: below

Vertical boundary: $x \geq 2$ is a **solid** vertical line; shade the **right**.

2 SYSTEMS: HOW MANY SOLUTIONS?

Graph	Algebra ends with	Solutions
lines cross	$x = 2, y = 3$	one point
parallel lines	$0 = 4$ (false)	none (inconsistent)
same line	$0 = 0$ (true)	infinitely many (dependent)

Example: Elimination: $3x + 2y = 12$ and $5x - 2y = 4$

Add (the y-terms cancel): $8x = 16$, so $x = 2$. Then $3(2) + 2y = 12$, $y = 3$. (2, 3)

Check: $5(2) - 2(3) = 4$ ✓

$2x - y = 3$
$4x - 2y = 10$
Row 2 − 2·(Row 1): **0 = 4** → no solution

$2x - y = 3$
$4x - 2y = 6$
Row 2 − 2·(Row 1): **0 = 0** → infinitely many

Multiply every term of an equation (right side too) to make opposites. **0 = 0 is not "no solution".**

3 SOLVING SYSTEMS IN CONTEXT

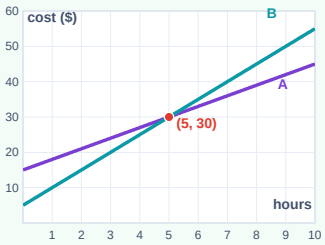
Example: Bike rental cost C after h hours

Shop A: $C = 15 + 3h$ **Shop B:** $C = 5 + 5h$

Substitution: $15 + 3h = 5 + 5h \rightarrow 10 = 2h \rightarrow h = 5$; $C = 30$.

(5 hours, $30)

Meaning: under 5 hours Shop B is cheaper; over 5 hours Shop A is cheaper. At 3 hours: A = $24, B = $20.



Substitution: $y = 2x - 1$ and $3x + y = 14$
$3x + (2x - 1) = 14 \rightarrow x = 3, y = 5$

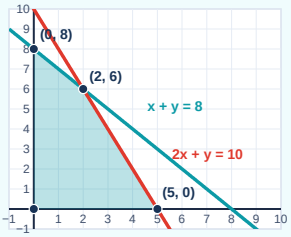
4 SYSTEMS OF INEQUALITIES & VERTICES

Example: $x + y \leq 8$, $2x + y \leq 10$, $x \geq 0$, $y \geq 0$

Graph each boundary, shade each side, and keep the **overlap** (the feasible region).

Vertices: intersect pairs of boundaries, then check each point in **all** inequalities.

x-axis & $2x + y = 10$: (5, 0)
$x + y = 8$ & $2x + y = 10$: subtract, $x = 2$, $y = 6$
y-axis & $x + y = 8$: (0, 8) axes: (0, 0)



The point (8, 0) is on $x + y = 8$ but fails $2x + y \leq 10$.

5 LINEAR PROGRAMMING

Example: Maximize $P = 30x + 20y$ on the region at left (corner points)

Vertex	(0, 0)	(5, 0)	(2, 6)	(0, 8)
P	0	150	**180**	160

Answer: the maximum is $P = 180$ at $(x, y) = (2, 6)$. Test **every** vertex; the best value is at a corner of the region, and a minimum works the same way (take the smallest).

In context: x = trays of muffins, y = trays of scones, P = profit in dollars: 2 trays of muffins and 6 of scones earn $180. Check: $2 + 6 = 8$ and $2(2) + 6 = 10$, so both limits are met.

If two vertices tie, every point on the edge between them ties too.

6 THREE EQUATIONS, THREE VARIABLES

Example: $x + y + z = 6$ (1) $2x - y + z = 3$ (2) $x + 2y - z = 2$ (3)

1. Eliminate the **same variable** twice: (1) + (3): $2x + 3y = 8$ (1) − (2): $-x + 2y = 3$

2. Solve the 2-variable system: $x = 2y - 3$, so $2(2y - 3) + 3y = 8$, $7y = 14$, $y = 2$, $x = 1$

3. Back-substitute in (1): $1 + 2 + z = 6$, $z = 3$. (1, 2, 3)

4. Check in (2) and (3): $2 - 2 + 3 = 3$ ✓ $1 + 4 - 3 = 2$ ✓

Contradiction (like $0 = 5$): no solution. **0 = 0:** infinitely many. Word problems: define x, y, z first, then write one equation per fact.

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Dividing an inequality by a **negative** flips the symbol.
- Dashed for < >, solid for ≤ ≥; test a point to choose the side.
- Multiply the **whole** equation when you eliminate.
- Check every vertex in the objective function.
- Check a solution in **all** original equations.

COMMON MISTAKES:

- Shading the wrong side or forgetting to flip the inequality.
- A sign slip when subtracting equations.
- Calling 0 = 0 "no solution" (it means infinitely many).
- Testing only some vertices in linear programming.
- Not checking a corner point against every constraint.